

**SentinelTask - Secure Task & Project
Management System with Intelligent Access
Control**

submitted to

VANITA VISHRAM WOMEN'S UNIVERSITY

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1. Introduction

1.1 Existing System

Small and mid-sized teams today rely on spreadsheets, email threads, and generic chat applications to track tasks and deadlines. Generic productivity tools such as Trello, Asana, and Notion offer boards and lists, but they lack an organization-specific role hierarchy — most apply the same permissions to every logged-in user, with no built-in Admin / Manager / Employee separation.

Password recovery and account security in many of these tools are often bolted on rather than designed around hashed credentials and time-bound tokens, and there is typically no unified activity trail linking who created, updated, or deleted a project or task.

1.2 Need for the New System

- Centralized Platform — one secure system for projects, tasks, team, calendar, files and reports instead of scattered tools.
- Role-Aware Access — admins, managers and employees each need a different view and a different set of permissions.
- Stronger Security — passwords must be hashed, sessions token-based, and password recovery time-bound and traceable.
- Accountability — every project should record who created it, so responsibility is never ambiguous.

1.3 Objective of the New System

- Build a secure authentication system using JWT and bcrypt password hashing.
- Implement Role-Based Access Control (RBAC) for three roles — Admin, Manager and Employee.
- Provide a dedicated dashboard and navigation experience for each role.
- Enable project management with status tracking (Planning → Active → Completed).
- Offer a secure, time-bound password reset workflow using hashed tokens.
- Design a modular architecture covering tasks, team, calendar, milestones, notifications, reports, activity log and files.
- Lay the foundation for an AI-assisted task prioritization module in a later phase.

1.4 Problem Definition

Small and mid-sized teams either use tools that are too generic — with flat, one-size-fits-all permissions — or systems that are too complex and costly to adopt. Neither approach enforces organizational hierarchy or keeps sensitive project data protected behind properly scoped, role-based access. This results in three recurring problems:

- Over-privileged access — any logged-in user can often see or edit data meant for a different role.
- No accountability trail — it is hard to tell who created or last touched a project.
- Weak recovery flow — password resets are frequently insecure or entirely manual.

1.5 Core Components

The system is organized into the following functional modules:

1. Authentication & Access Control
2. Admin / Manager / Employee Dashboards
3. Project Management
4. Task Management
5. Team Management
6. Calendar & Scheduling
7. Milestone Tracking
8. Notifications
9. Reports & Analytics
10. Activity Log
11. File Management
12. Settings

1.6 Project Profile

Project Title	SentinelTask — AI-Driven Secure Task & Project Management System with Intelligent Access Control
Category	Web Application — MERN Stack
Frontend	React 19, Vite, Tailwind CSS, React Router DOM
Backend	Node.js, Express.js
Database	MongoDB with Mongoose ODM
Security	JWT Authentication, bcrypt password hashing, Role-Based Access Control
Project Type	MSc.IT Final Year Major Project
Current Stage	Review — 2 (50% completion)

1.7 Assumptions and Constraints

Assumptions

- Users access the system through a modern web browser with an active internet connection.
- Each user is assigned exactly one role at a time — Admin, Manager or Employee.
- A MongoDB instance is available and reachable via the configured connection string.
- Email delivery for password reset will be integrated in a later production phase.

Constraints

- Fixed academic timeline — development is planned and reviewed in phases.
- Password reset currently runs in demo mode, returning the reset token directly instead of emailing it.

- Access control is role-based, not yet a granular per-permission ACL system.
- Several modules (Tasks, Team, Calendar, etc.) currently have complete UI but are pending database integration.

1.8 Advantages and Limitations of the Proposed System

Advantages

- Centralized, secure system replacing scattered spreadsheets and chat threads.
- Role-based dashboards give each user exactly the view and controls they need.
- Passwords are hashed with bcrypt; sessions use signed JWT tokens.
- Password reset tokens are hashed and expire in 15 minutes.
- Modular MERN architecture makes it straightforward to add new modules.

Limitations (Current Stage)

- AI-driven task prioritization is planned but not yet implemented.
- Most modules beyond Authentication and Projects are frontend-only, not yet wired to the database.
- No real email service integrated for password reset delivery yet.
- Automated test suite is not yet in place.

2. Requirement Determination Analysis

2.1 Requirement Determination

Functional Requirements

- Register and log in with validated name, email, password and role.
- Redirect each user to the dashboard for their role after login.
- Recover a forgotten password via a secure, time-bound token.
- Create, view, update and delete projects, restricted by role.
- Manage tasks, team, calendar, milestones, notifications, reports, activity log and files.

Non-Functional Requirements

- Security — JWT authentication, bcrypt hashing, hashed reset tokens.
- Performance — Vite-powered frontend, lightweight Express APIs.
- Usability — role-specific dashboards and sidebar navigation.
- Scalability — modular MERN structure supports new modules easily.
- Maintainability — routes, models and middleware kept separate.

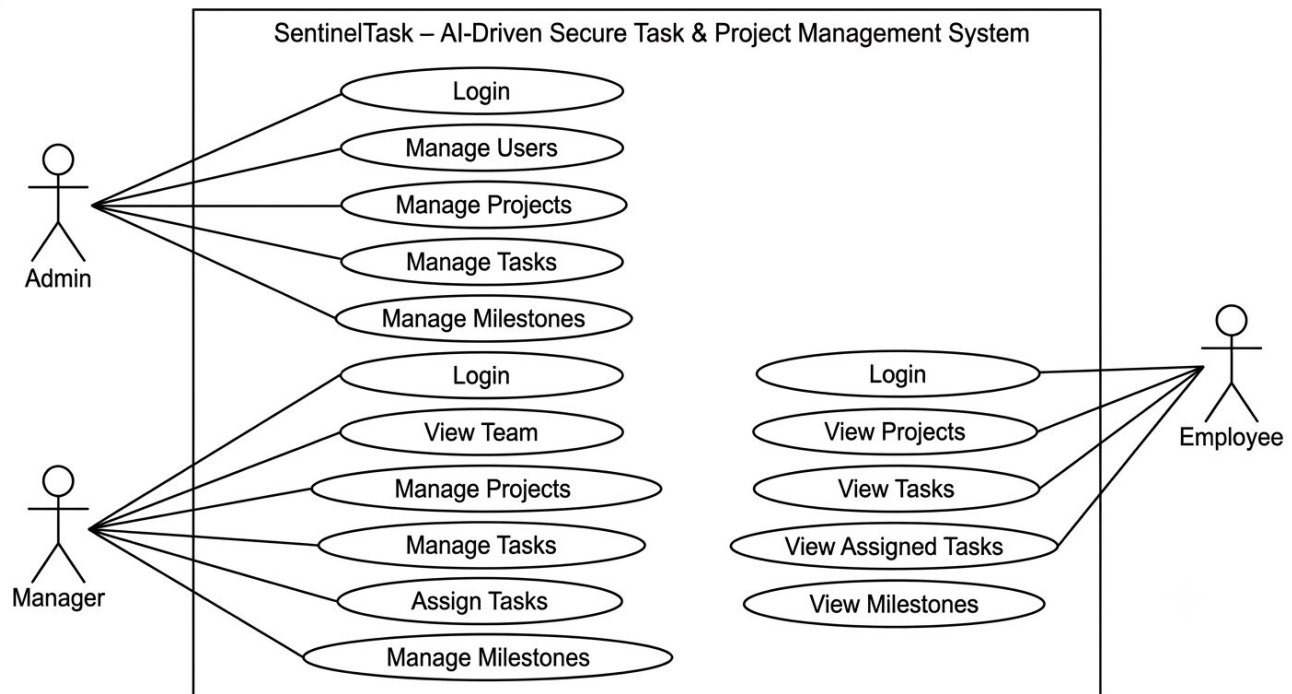
2.2 Targeted Users

Admin	Full system access — manage all projects, view admin-only routes, and the only role that can delete a project.
Manager	Create and manage projects and tasks, view team-level dashboards and reports.
Employee	View assigned tasks, update personal progress, and access a focused personal dashboard.

3. System Design

3.1 Use Case Diagram

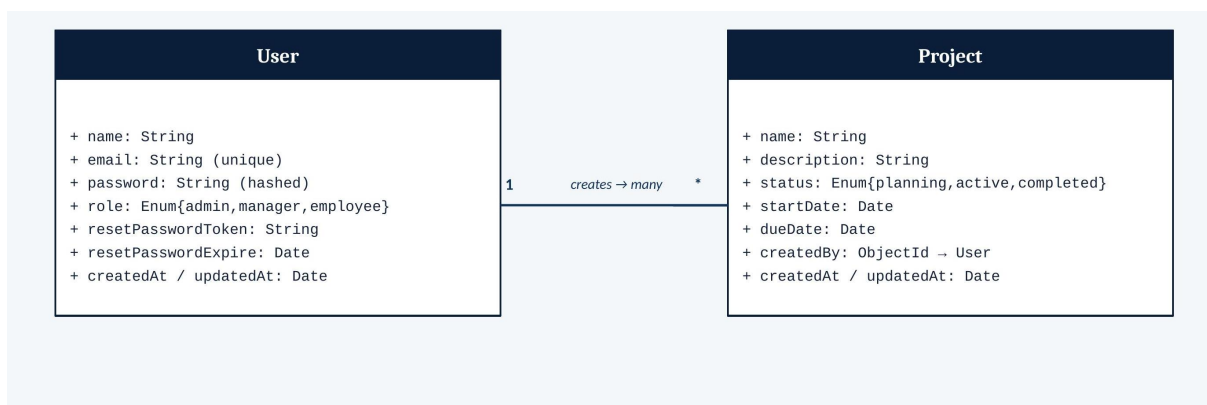
The diagram below shows the three actors — Admin, Manager and Employee — and the use cases available inside the SentinelTask system. Coloured dots on each use case indicate which role(s) can perform that action.



Use Case Diagram

3.2 Class Diagram

The class diagram reflects the two core Mongoose models in the current codebase — User and Project — and the relationship between them.

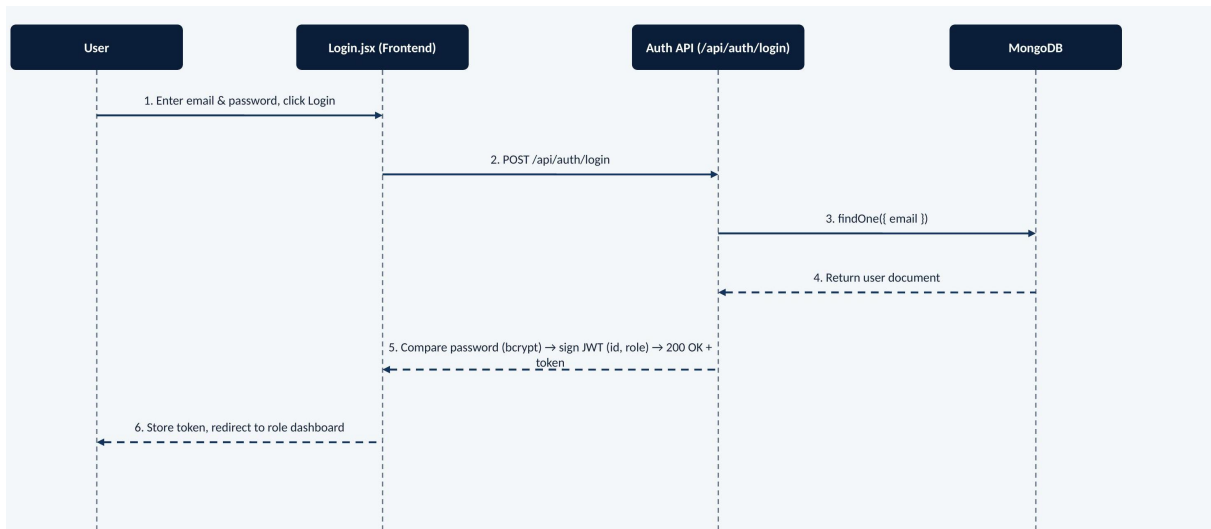


Class Diagram (User and Project models)

One User (with role Admin or Manager) can create many Projects. Each Project stores a reference back to its creator through the createdBy field.

3.3 Interaction Diagram

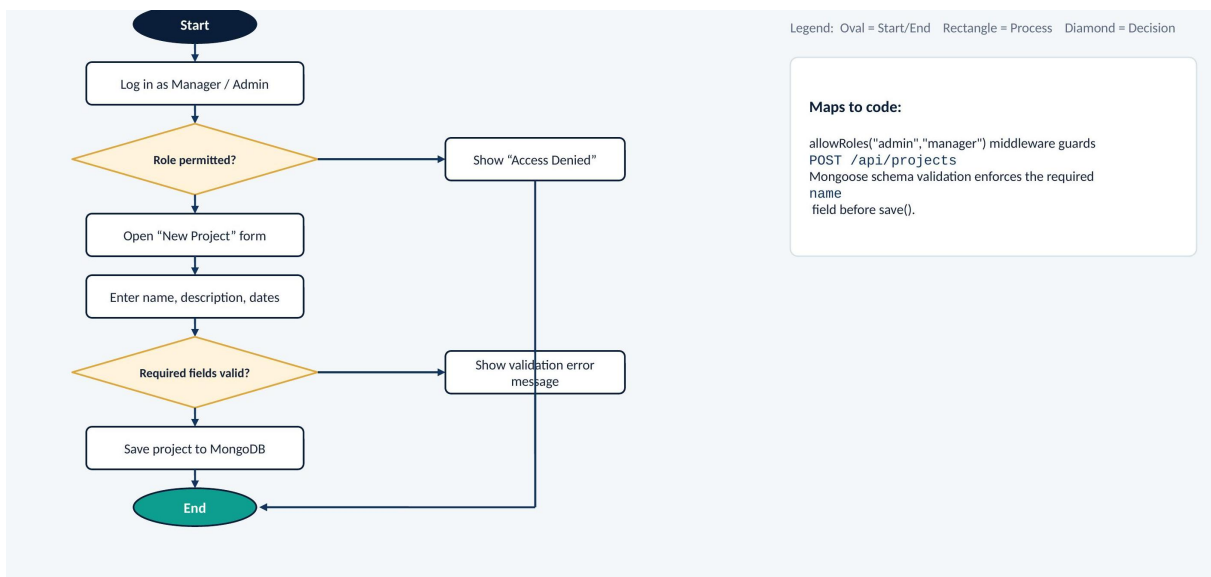
The sequence diagram below traces a login request from the user interface through to the database and back, matching the implementation in the authentication route.



Interaction (Sequence) Diagram: Login Flow

3.4 Activity Diagram

The activity diagram below models the workflow for creating a new project, including the role-permission check enforced by the allowRoles middleware and the field-level validation enforced by the Mongoose schema.



Activity Diagram: Create Project Workflow

3.5 Data Dictionary

User Collection

Field	Type	Constraints / Description
name	String	Required, 2–50 characters, letters and spaces only
email	String	Required, unique, must match a valid email format
password	String	Required, minimum 6 characters, stored as a bcrypt hash
role	Enum	admin manager employee (default: employee)
resetPasswordToken	String	SHA-256 hash of the reset token, set only during a reset request
resetPasswordExpire	Date	Token validity window — 15 minutes from issue
createdAt / updatedAt	Date	Auto-managed by Mongoose timestamps()

Project Collection

Field	Type	Constraints / Description
name	String	Required
description	String	Optional, defaults to an empty string
status	Enum	planning active completed (default: planning)
startDate	Date	Optional
dueDate	Date	Optional
createdBy	ObjectId	Required, references the User collection
createdAt / updatedAt	Date	Auto-managed by Mongoose timestamps()

Both collections include auto-managed timestamps — `createdAt` and `updatedAt` — provided by Mongoose's `{ timestamps: true }` schema option.

4. Development

4.1 Coding Standards

- Modular folder structure — routes, models, middleware and config kept in separate files.
- Consistent naming — camelCase for variables/functions, PascalCase for React components and Mongoose models.
- RESTful API design — /api/auth and /api/projects use proper HTTP verbs (GET, POST, PUT, DELETE).
- Centralized error handling — try/catch blocks with meaningful HTTP status codes (400, 401, 403, 404, 409, 500).
- Input validation at two levels — Mongoose schema rules plus manual regex/length checks in route handlers.
- Secrets kept out of code — JWT_SECRET, database URI and port loaded from a .env file via dotenv.
- Password security — bcrypt hashing (10 salt rounds); reset tokens are SHA-256 hashed before being stored.

4.2 Screen Shots

The wireframes below preview the screens implemented so far — Login, Register, Forgot/Reset Password, Project List, and the Admin and Employee dashboards. These will be replaced with live application screenshots before final submission.

1. Register Page

SentinelTask
Secure Workspace

CREATE ACCOUNT

Start managing your work smarter.

Create your SentinelTask account and access your secure workspace.

Secure role-based workspace

Join SentinelTask

Create Account

Enter your details to create your workspace account.

Full Name
Enter your full name

Email Address
Enter your email

Select Role
Employee

Password
Create password Show

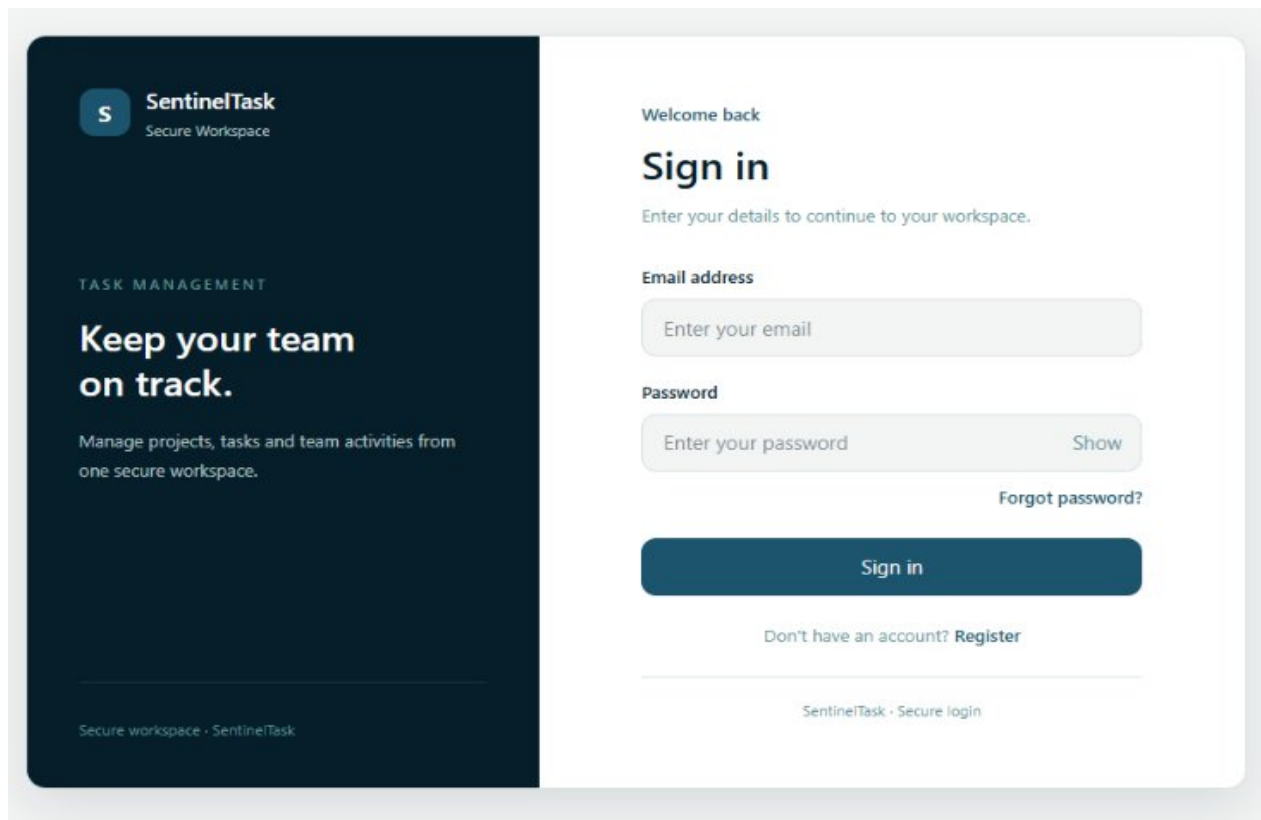
Confirm Password
Confirm your password Show

Create Account

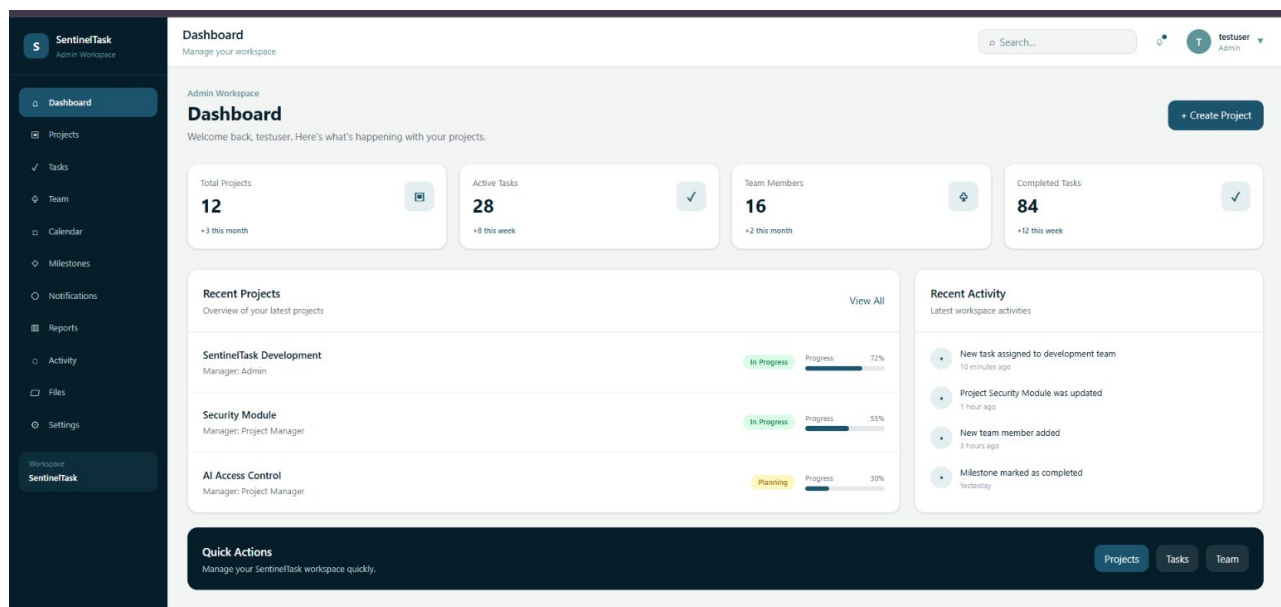
Already have an account? [Sign in](#)

SentinelTask - Secure role-based login

2. Sign in Page



3. Admin Panel



SentinelTask

Dashboard

WORKSPACE

Projects

Manage, monitor and organize all your projects.

New Project

Total Projects2

In Progress1

Completed1

Active Projects1

All Projects

View and manage your current projects.

2 Projects

F

fdghghnm

Created by: testuser

zxcghnmnbvbn,

Start Date9/17/2026

Due Date9/17/2026

Completed

ViewEditDelete

S

SentinelTask Project

Created by: Test Admin

Task management system for team

Start DateNot set

Due DateNot set

In Progress

ViewEditDelete

TASK MANAGEMENT

Tasks

+ New Task

Total Tasks3

To Do1

In Progress2

Completed0

Search tasks, projects or employees...

All Status

All Priority

All Tasks

View and manage your current tasks.

3 Tasks

S

swdfghju

Project: SentinelTask Project

xcdghyghjgjk

Assigned To: testtemp

Priority: Medium

Due Date: 9/18/2026

Status: To Do

To Do

ViewEditDelete

S

sdgrgt

Project: SentinelTask Project

dfghjklklghghfcdvbnm,jp87ytrfd

Assigned To: Unassigned

Priority: Medium

Due Date: 9/22/2026

Status: In Progress

In Progress

ViewEditDelete

Workspace

Team

Manage your team members, roles and workspace access.

+ Add Member

Total Members
5
All workspace members

Managers
2
Project managers

Employees
3
Workspace employees

Active Members
4
Currently active

Team Members

View and manage members of your SentinelTask workspace.

Showing 5 of 5 members

Search members...

All Roles

All Status

Member	Role	Department	Status	Action
RS Rahul Sharma rahul@sentineltask.com	Manager	Development	Active	Deactivate
PS Priya Shah priya@sentineltask.com	Manager	Design	Active	Deactivate
AP Amit Patel amit@sentineltask.com	Employee	Marketing	Active	Deactivate
NP Neha Patel neha@sentineltask.com	Employee	Development	Active	Deactivate
KM Karan Mehta	Employee	Testing	Inactive	Activate

SentinelTask

Dashboard

Workspace

Calendar

View and manage your tasks, meetings and important deadlines.

+ Add Event

September 2026
Select a date to view scheduled events.

< Today >

Sun	Mon	Tue	Wed	Thu	Fri	Sat
30	31	1 Homepage Design Review	2	3	4	5 1 Mobile UI Review
6	7 Login API Development	8 Marketing Campaign	9	10	11 Database Design	12 1
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	1	2	3
4	5	6	7	8	9	10

SELECTED DATE

Monday, September 7, 2026

+ No events
Nothing is scheduled for this date.
+ Add Event

Upcoming Events
Your next scheduled activities.

8 Login API Development
11:30 AM - Task

12 Mobile UI Review
02:00 PM - Meeting

15 Marketing Campaign
05:00 PM - Deadline

Workspace

Settings

Manage your account and workspace settings.

SETTINGS

Profile

Password

Notifications

Security

Appearance

Profile Settings

Manage your personal account information.

T

testuser
testuser123@gmail.com
Admin

Full Name

testuser

Email Address

testuser123@gmail.com

Role

Admin

Department

Management

Reset Changes

Save Changes

Workspace

Notifications

View and manage your latest notifications.

Mark All as ReadClear All

Total Notifications

6

All notifications

Unread

3

Need your attention

Read

3

Already viewed

Task Alerts

1

Task related updates

Search Notifications

Search notification...

Filter

All Notifications

Recent Notifications

6 notifications found

✓

New Task Assigned

TaskNew

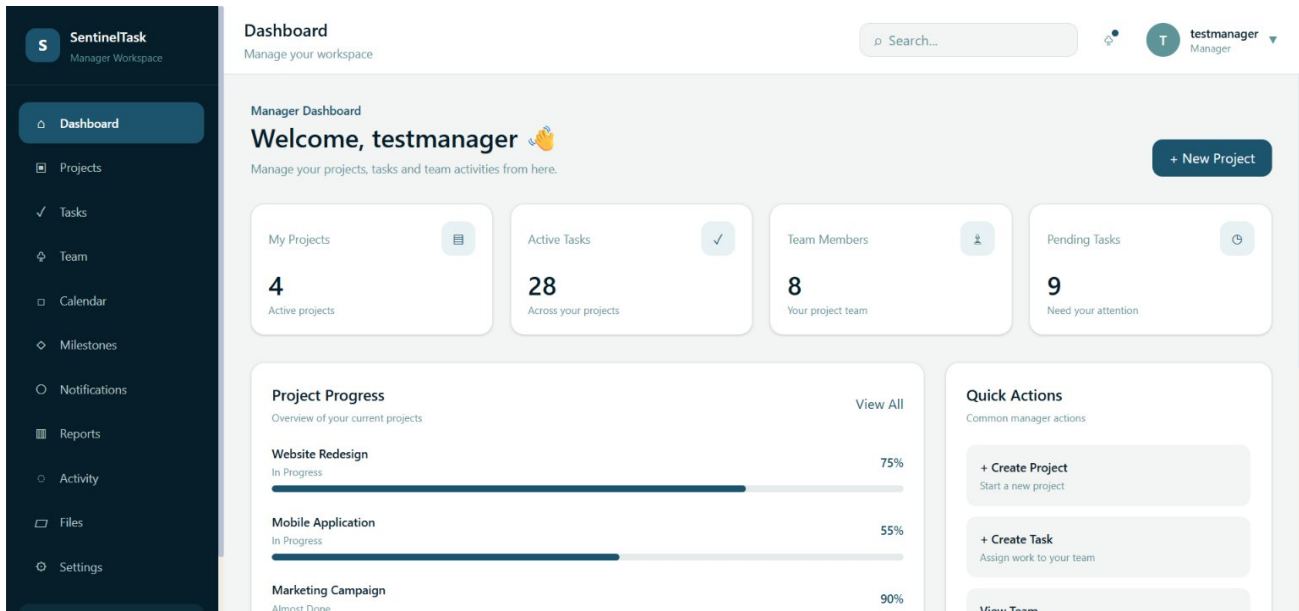
You have been assigned a new task: Design Homepage.

Mark as Read

Delete

10 minutes ago

4. ManagerPanel



The dashboard features a dark sidebar with navigation links: Dashboard, Projects, Tasks, Team, Calendar, Milestones, Notifications, Reports, Activity, Files, and Settings. The main content area is titled 'Dashboard' and 'Manager Dashboard'. It includes a search bar, a user profile for 'testmanager', and a '+ New Project' button. The dashboard displays four summary cards: 'My Projects' (4 Active projects), 'Active Tasks' (28 Across your projects), 'Team Members' (8 Your project team), and 'Pending Tasks' (9 Need your attention). Below these are two sections: 'Project Progress' showing progress bars for 'Website Redesign' (75%), 'Mobile Application' (55%), and 'Marketing Campaign' (90%); and 'Quick Actions' with buttons for '+ Create Project', '+ Create Task', and 'View Team'.

SentinelTask
Manager Workspace

Dashboard

Dashboard
Manage your workspace

Search...

testmanager
Manager

Manager Dashboard

Welcome, testmanager

Manage your projects, tasks and team activities from here.

+ New Project

My Projects
4
Active projects

Active Tasks
28
Across your projects

Team Members
8
Your project team

Pending Tasks
9
Need your attention

Project Progress
Overview of your current projects

View All

Website Redesign
In Progress
75%

Mobile Application
In Progress
55%

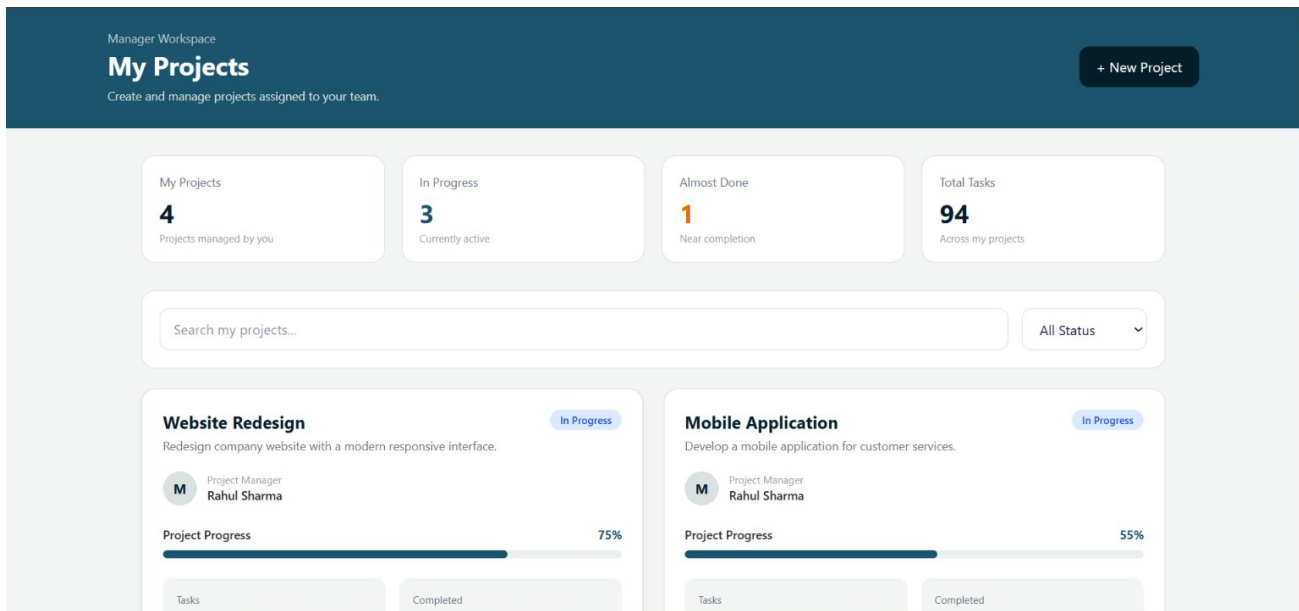
Marketing Campaign
Almost Done
90%

Quick Actions
Common manager actions

+ Create Project
Start a new project

+ Create Task
Assign work to your team

View Team



The 'My Projects' page has a dark header with 'Manager Workspace' and 'My Projects' (Create and manage projects assigned to your team.), along with a '+ New Project' button. It features four summary cards: 'My Projects' (4 Projects managed by you), 'In Progress' (3 Currently active), 'Almost Done' (1 Near completion), and 'Total Tasks' (94 Across my projects). Below these is a search bar 'Search my projects...' and a status filter 'All Status'. The main content shows two project cards: 'Website Redesign' (In Progress, 75% complete) and 'Mobile Application' (In Progress, 55% complete). Each card lists the project manager 'Rahul Sharma' and includes 'Tasks' and 'Completed' sub-sections.

Manager Workspace

My Projects
Create and manage projects assigned to your team.

+ New Project

My Projects
4
Projects managed by you

In Progress
3
Currently active

Almost Done
1
Near completion

Total Tasks
94
Across my projects

Search my projects...

All Status

Website Redesign
Redesign company website with a modern responsive interface.

Mobile Application
Develop a mobile application for customer services.

Project Progress
75%

Project Progress
55%

Tasks Completed

Tasks Completed

Manager Workspace

Manage Tasks

Create, assign and monitor tasks for your team.

+ Create Task

Total Tasks
5
Across your projects

To Do
2
Waiting to start

In Progress
2
Currently active

Completed
1
Finished tasks

Search tasks...

All Status

All Priority

Reset Filters

Design Homepage High Priority In Progress

Create responsive homepage design for the website.

Project
Website Redesign

Assigned To
Amit Patel

Due Date
2026-09-10

View

Edit

Complete

Manager Workspace

My Team

Manage your project team members and their assignments.

+ Add Team Member

Total Members
6
Team members

Active Members
5
Currently active

Total Tasks
36
Assigned to team

Completed Tasks
25
Successfully completed

Team Members
Members working under your projects

Search team member...

All

TEAM MEMBER	ROLE	DEPARTMENT	PROJECT	TASKS	STATUS	ACTIONS
A Amit Patel amit@example.com	Developer	Development	Website Redesign	6/8	Active	View Edit
N Neha Patel neha@example.com	Backend Developer	Development	ERP Development	4/7	Active	View Edit
K Karan Mehta karan@example.com	UI/UX Designer	Design	Mobile Application	4/6	Active	View Edit

SentinelTask

Manager Calendar

Dashboard

Calendar

Manage your team meetings, deadlines and project events.

+ Add Event

Total Events

5

Meetings

2

Deadlines

1

Project Events

5

Search events, projects or team members...

All Event Types

September 2026

Today

Sun	Mon	Tue	Wed	Thu	Fri	Sat
		1	2	3	4	5

Selected Date

8 September 2026

Meeting

Homepage Design Review

10:00 AM

Website Redesign

Amit Patel

View

SentinelTask

Manager Files

Dashboard

Files

Manage project documents, images and shared files.

+ Upload File

Total Files

8

All uploaded files

Documents

4

PDF, Word, Excel and more

Images

3

Project images and designs

Other Files

1

Other file formats

Search files...

All Files

Project Files

8 files found

D	Homepage UI Design.pdf	Document	Size: 2.4 MB	Uploaded by: Rahul Mehta	Project: Website Redesign	Date: 2026-09-07	View	Download	Delete
I	Mobile App Wireframe.png	Image	Size: 1.8 MB	Uploaded by: Priya Shah	Project: Mobile App Development	Date: 2026-09-06	View	Download	Delete

Workspace

Manager Settings
Manage your account and manager settings.

Dashboard

SETTINGS

Profile

Password

Notifications

Security

Appearance

Profile Settings
Manage your personal account information.

T

testmanager
testmanager123@gmail.com
Manager

Full Name

testmanager

Email Address

testmanager123@gmail.com

Role

Manager

Department

Management

Reset Changes

Save Changes

SentinelTask

Manager Notifications

Dashboard

Notifications
Stay updated with your projects, tasks and team activities.

Mark All as Read

Total Notifications
6

Unread
3

Read
3

Search notifications...

All Notifications

Recent Notifications
6 notifications found

T

Task

New Task Assigned
You have been assigned a new task for the Website Redesign project.
10 minutes ago

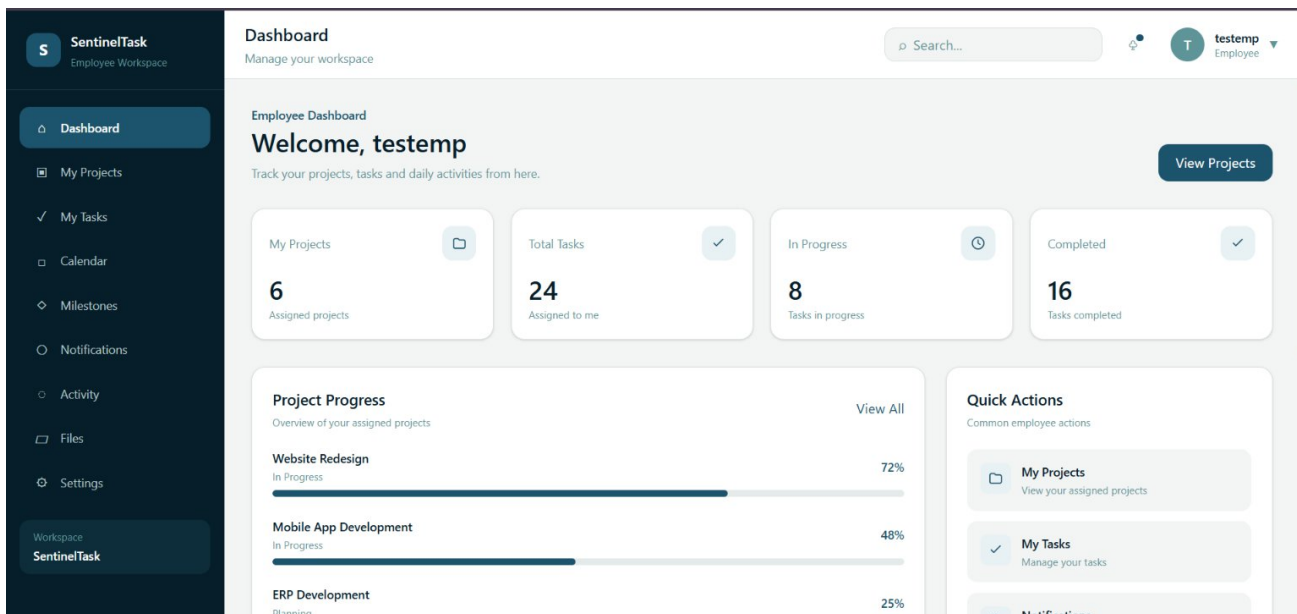
View Read Delete

I

Deadline

View Read Delete

5. Employee Panel



The screenshot shows the 'Employee Dashboard' in SentinelTask. The left sidebar contains navigation links: Dashboard, My Projects, My Tasks, Calendar, Milestones, Notifications, Activity, Files, and Settings. The main content area is titled 'Welcome, testemp' and includes a 'View Projects' button. It features four summary cards: 'My Projects' (6 Assigned projects), 'Total Tasks' (24 Assigned to me), 'In Progress' (8 Tasks in progress), and 'Completed' (16 Tasks completed). Below these is a 'Project Progress' section with a 'View All' link, showing progress bars for 'Website Redesign' (72%), 'Mobile App Development' (48%), and 'ERP Development' (25%). A 'Quick Actions' section on the right lists 'My Projects', 'My Tasks', and 'Notifications'.

SentinelTask
Employee Workspace

Dashboard
Manage your workspace

Search...

testemp
Employee

Employee Dashboard
Welcome, testemp
Track your projects, tasks and daily activities from here.

My Projects
6
Assigned projects

Total Tasks
24
Assigned to me

In Progress
8
Tasks in progress

Completed
16
Tasks completed

Project Progress
Overview of your assigned projects
View All

Website Redesign
In Progress
72%

Mobile App Development
In Progress
48%

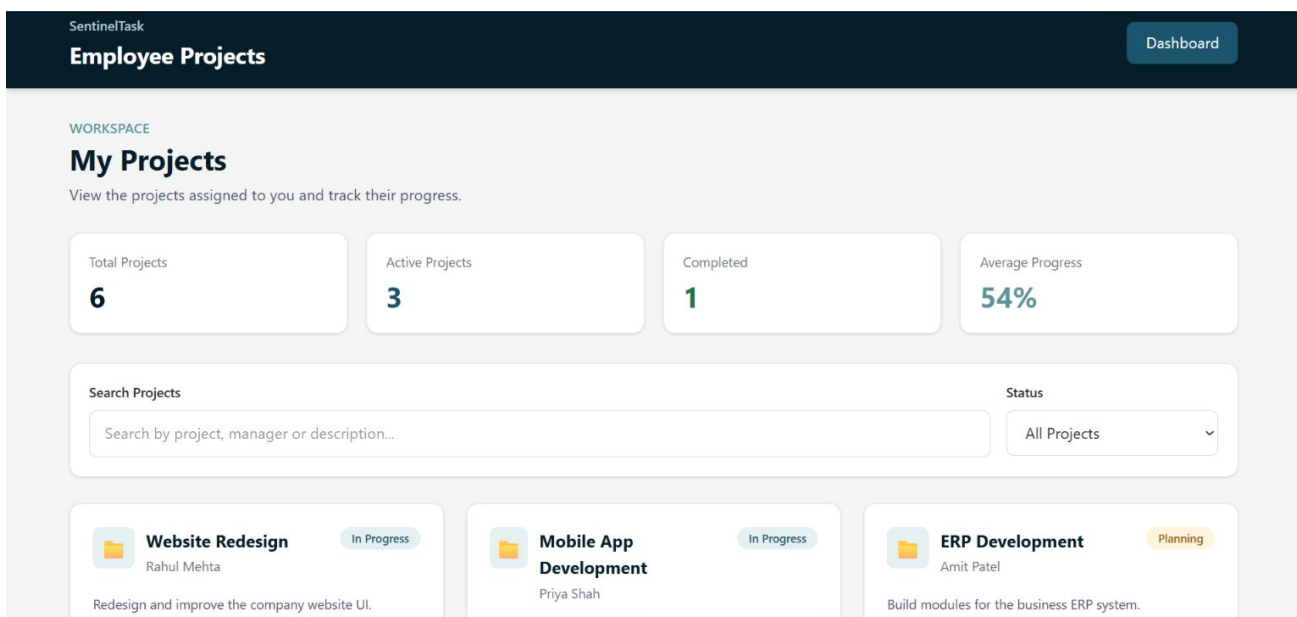
ERP Development
Planning
25%

Quick Actions
Common employee actions

My Projects
View your assigned projects

My Tasks
Manage your tasks

Notifications



The screenshot shows the 'Employee Projects' page in SentinelTask. The top header includes 'SentinelTask' and 'Employee Projects' with a 'Dashboard' button. The main content area is titled 'My Projects' and includes a 'View the projects assigned to you and track their progress.' message. It features four summary cards: 'Total Projects' (6), 'Active Projects' (3), 'Completed' (1), and 'Average Progress' (54%). Below these is a 'Search Projects' section with a search bar and a 'Status' dropdown menu. The bottom section displays three project cards: 'Website Redesign' (In Progress, Rahul Mehta), 'Mobile App Development' (In Progress, Priya Shah), and 'ERP Development' (Planning, Amit Patel).

SentinelTask
Employee Projects
Dashboard

My Projects
View the projects assigned to you and track their progress.

Total Projects
6

Active Projects
3

Completed
1

Average Progress
54%

Search Projects
Search by project, manager or description...

Status
All Projects

Website Redesign
Rahul Mehta
In Progress
Redesign and improve the company website UI.

Mobile App Development
Priya Shah
In Progress

ERP Development
Amit Patel
Planning
Build modules for the business ERP system.

SentinelTask

Employee Tasks

Dashboard

WORKSPACE

My Tasks

View your assigned tasks and keep track of your work.

Total Tasks

6

Pending

2

In Progress

2

Completed

2

Search Tasks

Status

Priority

Search task, project or manager...

All Status

All Priority

 **Create Homepage UI** In Progress High

Design the responsive homepage for the company website.

View Complete

Website Redesign Rahul Mehta Due: 10 Sep 2026

Employee Workspace

My Files

Dashboard

E

Workspace Files

My Files

Access and manage files related to your assigned projects.

Upload File

Total Files

6

Available files

Documents

2

Project documents

Reports

1

Project reports

Projects

6

Projects with files

Project Files

Files shared with your projects

Search files...

All Document Design Report Archive Spreadsheet

 **Homepage UI Design.fig**

Website Redesign • 2.4 MB

Design

Uploaded by **Rahul Mehta**
06 Sep 2026

View Download Delete

EMPLOYEE WORKSPACE

My Calendar

Dashboard

E

Employee Calendar

View your tasks, meetings and project milestones

Total Events
8
Scheduled events

Tasks
4
Task deadlines

Meetings & Milestones
4
Important events

< September 2026 >

Today

Sun	Mon	Tue	Wed	Thu	Fri	Sat
		1	2	3	4	5
6	7	8	9	10 1 event Homepage UI Dea...	11	12 2 events Mobile App Team ... Login Validation D

Workspace

Employee Settings

Manage your account and employee settings.

Dashboard

SETTINGS

Profile

Password

Notifications

Security

Appearance

Profile Settings

Manage your personal account information.

T

testemp
testemp123@gmail.com
Employee

Full Name

testemp

Email Address

testemp123@gmail.com

Role

Employee

Department

Development

Reset Changes

Save Changes